

For half wave rectified circuit, $V_{rms} = V_0 / 2 = 5 \text{ V}$
 $I_{rms} = V_{rms} / R = 5 / 50 = 0.10 \text{ A}$

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B

Using Einstein's Photoelectric equation, $\frac{hc}{\lambda} = \phi + eV_s$

$$\begin{aligned}\phi &= \frac{hc}{\lambda} - eV_s \\ &= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{150 \times 10^{-9}} - (1.60 \times 10^{-19})(2.1) \\ &= 9.9 \times 10^{-19} \text{ J} \\ &= 6.2 \text{ eV}\end{aligned}$$

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C

The ground state hydrogen atoms have electrons starting at E_1 .